Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	18
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3243.94	2186.5	2760.08	px^2
major_axis_length_px	lacuna shape	90.173	76.7457	32.1955	px
minor_axis_length_px	lacuna shape	48.2901	46.9029	21.1993	px
aspect_ratio	lacuna shape	1.9944	1.9215	0.5799	unitless
eccentricity	lacuna shape	0.8225	0.8532	0.1074	unitless
solidity	lacuna shape	0.745	0.7315	0.0977	unitless
roots_count	headline	12.3333	10.5	5.8107	count
ring_length_r30_px	headline	330.389	331.5	137.515	px
ring_length_r60_px	reported	675	663	290.5	px
owned_length_px	ownership-dependent	870.219	875.403	690.454	px
edge_count	ownership-dependent	42.8333	41	33.2199	count
mean_edge_length_px	ownership-dependent	20.8382	19.4081	6.1178	px
perimeter_px	normalised	298.015	280.756	117.796	px
ring_area_r30_px2	normalised	10062.7	9026	3149.84	px^2
ring_area_r60_px2	normalised	21478.1	20564	5976.9	px^2
in_frame_fraction_r30	normalised	0.9849	1	0.0369	unitless
in_frame_fraction_r60	normalised	0.9679	1	0.0742	unitless
ring_density_r30	normalised	0.0326306	0.0324353	0.00803934	px^-1
ring_density_r60	normalised	0.0311983	0.0342356	0.00899995	px^-1
roots_per_100px_perimeter	normalised	4.1092	4.1609	0.8051	per 100 px
ring_attached_length_r30_px	network v2	304.722	305.5	140.931	px
ring_attached_length_r60_px	network v2	530.833	569	312.814	px
ring_length_w_r30_px	network v2	375.612	380.2	155.075	px
ring_length_w_r60_px	network v2	766.47	768.796	330.615	px
sholl_crossings_r10	network v2	7.3333	6	3.4641	count
sholl_crossings_r20	network v2	7.9444	8	4.5045	count
sholl_crossings_r30	network v2	9.2778	8.5	4.2951	count
loop_count	loops, report only	2.4444	1	3.9738	count
loop_edge_share	loops, report only	0.1294	0.0594	0.1655	share
ring_width_mean_r30_px	width	7.6045	7.6553	0.6127	px

Per field
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measure	value	units
analysed_area_px2	977417	px^2
total_skeleton_length_px	25242	px
canalicular_length_density_per_px	0.0258252	px^-1
junction_count	540	count
junction_density_per_px2	0.00055248	px^-2
lacuna_count	21	count

lacunae_per_px2	2.149e-05	px^-2
skeleton_component_count	344	count
mean_component_length_px	73.3779	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0292605	px^-1
field_density_without_flagged_per_px	0.0283051	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	196	px
unattached_component_count	12	count
canalicular_length_density_attached_per_px	0.0258252	px^-1
field_length_density_with_unattached_per_px	0.0260257	px^-1
loop_count	56	count
loop_edge_share	0.1493	share
edge_count	1380	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2167	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23437	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
3	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
4	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
5	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
6	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
7	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
8	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
9	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
10	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
11	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
12	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
13	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
14	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
15	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
16	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
17	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
18	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
19	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False
20	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
21	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	395	566	913.767	51	17.917	6.7969
2	9	352	597	847.37	50	16.9474	7.3539
3	10	291	689	153.024	13	11.7711	6.7795
4	15	438	737	1092.65	58	18.8388	7.5509
5	8	322	834	958.808	50	19.1762	7.7452
6	6	155	358	401.421	19	21.1274	8.7355
7	26	659	1235	2350.11	115	20.4357	8.1509
8	16	500	1183	2498.2	117	21.3522	8.5243
9	11	341	658	1054.25	55	19.1681	8.3777
10	8	192	330	229.007	10	22.9007	6.0936
11	12	360	668	1001.49	35	28.614	7.7108
12	5	155	258	175.125	10	17.5125	8.2917
13	19	351	508	723.027	34	21.2655	5.9354
14	19	385	862	1396.18	70	19.9455	7.8658
15	14	408	1045	1085.1	51	21.2766	7.1052
16	15	277	585	661.831	36	18.3842	7.6247
17	8	167	425	202.48	5	40.496	7.0847
18	23	499	869	903.436	46	19.6399	7.5211
19	9	206	348	261.953	15	17.4636	7.6859
20	8	232	486	635.458	25	25.4183	8.1991
21	6	163	341	157.468	10	15.7468	7.2164

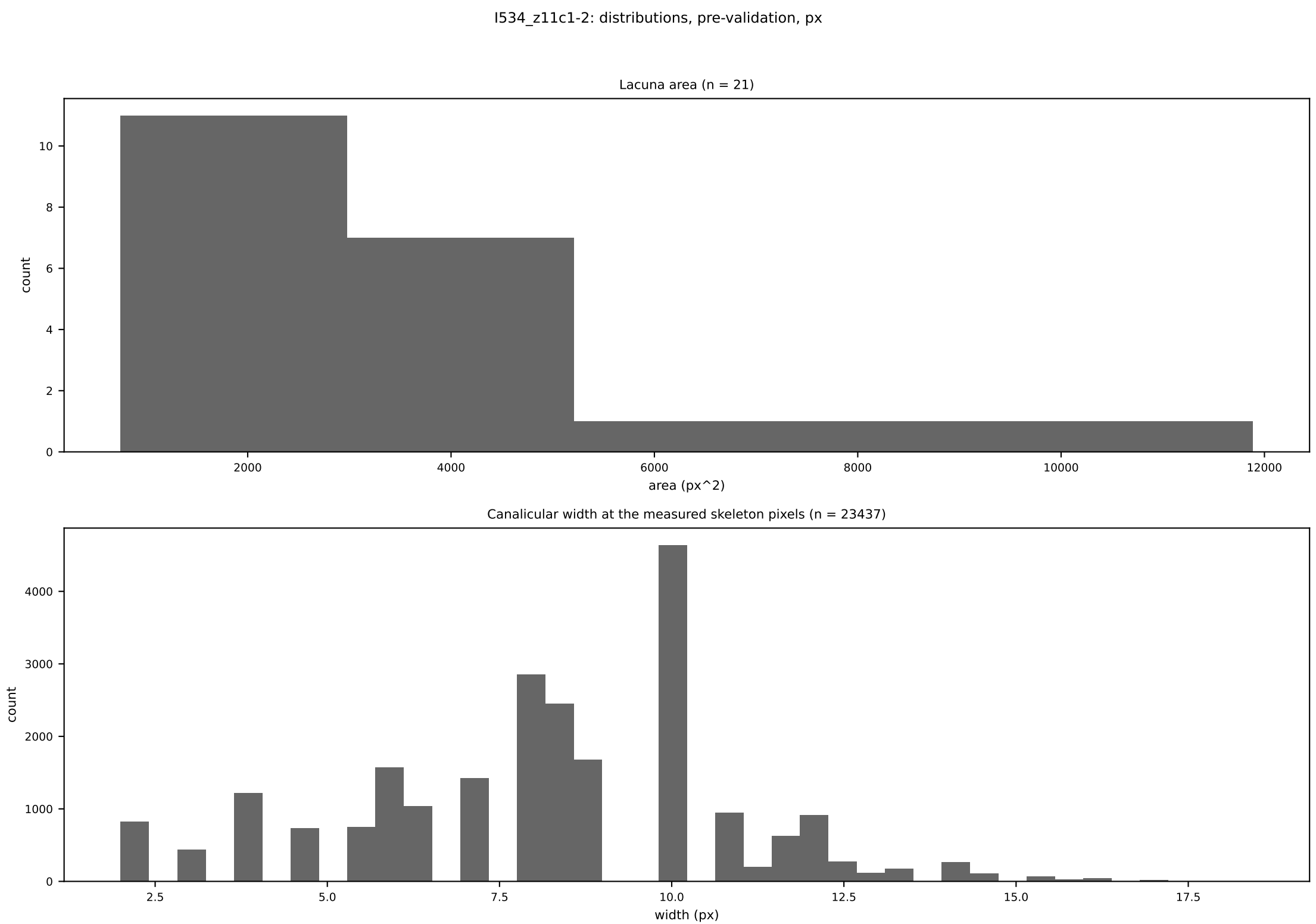
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	9574	14680	0.7421	0.6708	0.0412576	0.0385559	3.0343
2	270.635	8169	16049	0.8695	0.7737	0.0430897	0.0371986	3.3255
3	236.936	8094	17976	1	1	0.0359526	0.0383289	4.2205
4	341.505	11247	24451	1	1	0.0389437	0.0301419	4.3923
5	176.167	7690	20930	1	1	0.0418726	0.0398471	4.5412
6	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
7	548.257	17191	33534	1	1	0.038334	0.0368283	4.7423
8	297.706	11030	25469	1	1	0.0453309	0.0464486	5.3744
9	305.605	10599	17028	1	1	0.0321729	0.0386422	3.5994
10	191.167	7524	16666	0.9189	0.7962	0.0255183	0.0198008	4.1848
11	238.451	9266	19320	1	1	0.0388517	0.0345756	5.0325
12	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
13	490.399	11727	20563	0.8061	0.7376	0.0299309	0.0247046	3.8744
14	378.475	12733	27257	1	1	0.0302364	0.0316249	5.0201
15	373.546	12922	30830	1	1	0.0315741	0.0338956	3.7479
16	290.877	7526	16187	1	1	0.0368057	0.0361401	5.1568
17	233.38	6504	15282	1	1	0.0256765	0.0278105	3.4279
18	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
19	383.889	12894	27695	1	1	0.0159764	0.0125655	2.3444
20	222.693	8786	20198	1	0.9917	0.0264056	0.0240618	3.5924
21	196.066	8128	21509	1	1	0.0200541	0.0158538	3.0602

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	376	544	444.848	632.831	10	10	7
2	348	545	392.836	660.688	6	9	6
3	193	193	334.291	781.826	7	3	8
4	418	648	503.274	838.938	11	11	13
5	284	643	367.563	955.779	6	11	10

6	140	252	165.669	393.836	3	4	4
7	639	1187	726.487	1384.67	15	15	17
8	486	1110	568.688	1354.13	10	16	18
9	327	612	395.505	761.553	8	9	12
10	148	172	222.936	387.103	5	2	5
11	340	593	412.605	776.039	6	7	9
12	145	178	175.297	287.551	4	4	4
13	327	352	403.576	592.541	7	3	5
14	376	722	438.233	975.264	11	10	11
15	395	767	467.961	1172.69	8	11	12
16	232	384	316.179	665.186	3	6	7
17	151	181	182.681	476.274	6	2	5
18	451	748	578.399	1005.9	13	14	13
19	198	278	225.024	380.421	5	4	7
20	232	432	267.794	552.902	6	6	7
21	122	162	185.267	379.534	2	3	3


I534_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 21)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 23437)
count
width (px)